

1. System Workflow

The operational sequence of the FitBuddy application follows a structured workflow:

- 1. **Initiation:** User starts the web server using `streamlit run app.py` and accesses the UI on localhost.
- 2. **Data Capture:** User inputs numeric attributes (Age, Weight, Height) and chooses dropdown options (Diet, Goal).
- 3. **Trigger Event:** User clicks the "Generate Fitness Plan" button.
- 4. **Evaluation & Processing:** The Streamlit execution engine executes conditional rules based on the user's input state.
- 5. **Display:** The system renders the tailored workout schedule and customized meal routine.

2. Data Flow Diagram (DFD Level 0)

Source / Entity	Input Data Stream	Processing Unit	Output Data Stream	Destination
End User	Age, Weight, Height, Diet Preference, Goal	FitBuddy Engine (Streamlit Application)	Tailored Workout Routine & Diet Plan	User Browser Display